

PSO vs DQN - Stat Tests

Aheer Sravon

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1 Imports

```
[1]: import numpy as np
import pandas as pd
from scipy.stats import permutation_test
import itertools
```

2 Data Loading

```
[2]: models = [
    "logit", # row 0
    "hotelling", # row 1
    "linear" # row 2
]

firm1 = "pso"
firm2 = "dqn"

shock_none_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_a_data = {
```

```

model: {
  "avg_prices": {
    firm1: [],
    firm2: []
  },
  "deltas": {
    firm1: [],
    firm2: []
  },
  "rpdis": {
    firm1: [],
    firm2: []
  }
} for model in models
}

shock_b_data = {
  model: {
    "avg_prices": {
      firm1: [],
      firm2: []
    },
    "deltas": {
      firm1: [],
      firm2: []
    },
    "rpdis": {
      firm1: [],
      firm2: []
    }
  }
  for model in models
}

shock_c_data = {
  model: {
    "avg_prices": {
      firm1: [],
      firm2: []
    },
    "deltas": {
      firm1: [],
      firm2: []
    },
    "rpdis": {
      firm1: [],
      firm2: []
    }
  }
}

```

```

    } for model in models
}

for i in range(1,8):
    shock_none_df = pd.read_csv(f"../results/run_{i}/shock_none/
↪{firm1}_vs_{firm2}.csv")
    shock_a_df = pd.read_csv(f"../results/run_{i}/shock_a/
↪{firm1}_vs_{firm2}_schemeA.csv")
    shock_b_df = pd.read_csv(f"../results/run_{i}/shock_b/
↪{firm1}_vs_{firm2}_schemeB.csv")
    shock_c_df = pd.read_csv(f"../results/run_{i}/shock_c/
↪{firm1}_vs_{firm2}_schemeC.csv")

    # Map each dataframe to its corresponding data storage dictionary
    shock_mappings = [
        (shock_none_df, shock_none_data),
        (shock_a_df, shock_a_data),
        (shock_b_df, shock_b_data),
        (shock_c_df, shock_c_data)
    ]

    for df, data_dict in shock_mappings:
        for idx, model in enumerate(models):
            # Access the row matching the current model by its index (0, 1, or
↪2)

            row = df.iloc[idx]

            # Dynamically look up columns based on firm name capitalization (e.
↪g., "Q" and "PSQ")
            f1_upper = firm1.upper()
            f2_upper = firm2.upper()

            # Append Average Prices
            data_dict[model]["avg_prices"][firm1].append(row[f"{f1_upper} Avg.
↪Prices"])
            data_dict[model]["avg_prices"][firm2].append(row[f"{f2_upper} Avg.
↪Prices"])

            # Append Extra-profits Δ (deltas)
            data_dict[model]["deltas"][firm1].append(row[f"{f1_upper}
↪Extra-profits Δ"])
            data_dict[model]["deltas"][firm2].append(row[f"{f2_upper}
↪Extra-profits Δ"])

            # Append RPDI
            data_dict[model]["rpdis"][firm1].append(row[f"{f1_upper} RPDI"])

```

```
data_dict[model]["rpdis"][firm2].append(row[f"{f2_upper} RPDI"])
```

```
[3]: def paired_mean_diff(x, y, axis=0):
      return np.mean(x - y, axis=axis)
```

3 Logit

3.1 Shock None

3.1.1 Average Prices

```
[4]: array1 = np.array(shock_none_data['logit']['avg_prices'][firm1])
      array2 = np.array(shock_none_data['logit']['avg_prices'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)
```

```

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0100
Exact P-value:           0.01562
95% Test-Inverted CI:     [0.0100, 0.0100]
=====

```

3.1.2 Deltas

```

[5]: array1 = np.array(shock_none_data['logit']['deltas'][firm1])
     array2 = np.array(shock_none_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

```

```

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:    {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.1257
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.1200, 0.1300]
=====

```

3.1.3 RPDIs

```
[6]: array1 = np.array(shock_none_data['logit']['rpdis'][firm1])
array2 = np.array(shock_none_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the_
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
```

```
# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")
```

```
=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0129
Exact P-value:           0.01562
95% Test-Inverted CI:     [0.0100, 0.0200]
=====
```

3.2 Shock A

3.2.1 Average Prices

```
[7]: array1 = np.array(shock_a_data['logit']['avg_prices'][firm1])
array2 = np.array(shock_a_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
```

```

        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the
    ↪ CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0000
Exact P-value:           1.00000
95% Test-Inverted CI:     [0.0000, 0.0000]
=====

```

3.2.2 Deltas

```

[8]: array1 = np.array(shock_a_data['logit']['deltas'][firm1])
     array2 = np.array(shock_a_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪ dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

```

```

)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    ↪CI
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
EXACT PAIRED PERMUTATION TEST REPORT
=====

```

```

Observed Mean Difference: 0.4986
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.4850, 0.5000]
=====

```

3.2.3 RPDIs

```

[9]: array1 = np.array(shock_a_data['logit']['rpdis'][firm1])
     array2 = np.array(shock_a_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)

```

```

        ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
    else:
        ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0000
Exact P-value:           1.00000
95% Test-Inverted CI:    [0.0000, 0.0000]
=====

```

3.3 Shock B

3.3.1 Average Prices

```

[10]: array1 = np.array(shock_b_data['logit']['avg_prices'][firm1])
      array2 = np.array(shock_b_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

```

```

perm_res = permutation_test(
    data=(shifted_array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',
    alternative='two-sided',
    n_resamples=np.inf
)

# If p > 0.05, this shift magnitude is statistically plausible (inside the
↪ CI)
if perm_res.pvalue > 0.05:
    ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0043
Exact P-value:           0.25000
95% Test-Inverted CI:     [0.0000, 0.0100]
=====

```

3.3.2 Deltas

```

[11]: array1 = np.array(shock_b_data['logit']['deltas'][firm1])
      array2 = np.array(shock_b_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,

```

```

    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```
=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.1671
Exact P-value:          0.01562
95% Test-Inverted CI:   [0.1500, 0.1800]
=====
```

3.3.3 RPDIs

```
[12]: array1 = np.array(shock_b_data['logit']['rpdis'][firm1])
      array2 = np.array(shock_b_data['logit']['rpdis'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
```

```

if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0114
Exact P-value:           0.01562
95% Test-Inverted CI:     [0.0100, 0.0150]
=====

```

3.4 Shock C

3.4.1 Average Prices

```

[13]: array1 = np.array(shock_c_data['logit']['avg_prices'][firm1])
      array2 = np.array(shock_c_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

```

```

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0000
Exact P-value:           1.00000
95% Test-Inverted CI:     [0.0000, 0.0000]
=====

```

3.4.2 Deltas

```
[14]: array1 = np.array(shock_c_data['logit']['deltas'][firm1])
      array2 = np.array(shock_c_data['logit']['deltas'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
      if len(ci_values) > 0:
          lower_bound = np.min(ci_values)
          upper_bound = np.max(ci_values)
          ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
      else:
          ci_str = "Outside defined search grid"
```

```
# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")
```

```
=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.6871
Exact P-value:           0.01562
95% Test-Inverted CI:     [0.4550, 0.5000]
=====
```

3.4.3 RPDIs

```
[15]: array1 = np.array(shock_c_data['logit']['rpdis'][firm1])
      array2 = np.array(shock_c_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
```

```

        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:      {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0100
Exact P-value:           0.06250
95% Test-Inverted CI:      [-0.0200, 0.0000]
=====

```

4 Hotelling

4.1 Shock None

4.1.1 Average Prices

```

[16]: array1 = np.array(shock_none_data['hotelling']['avg_prices'][firm1])
      array2 = np.array(shock_none_data['hotelling']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies

```

```

        alternative='two-sided',
        n_resamples=np.inf
    )

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
# 05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    # CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```
array1 = array([1.14, 1.14, 1.14, 1.14, 1.14, 1.14, 1.14]), array2 =
array([1.12, 1.12, 1.12, 1.12, 1.12, 1.12, 1.12])
```

```
=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0200
Exact P-value:          0.01562
95% Test-Inverted CI:   Outside defined search grid
=====
```

4.1.2 Deltas

```
[17]: array1 = np.array(shock_none_data['hotelling']['deltas'][firm1])
array2 = np.array(shock_none_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the_
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)
```

```

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0643
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.0700, -0.0600]
=====

```

4.1.3 RPDIs

```

[41]: array1 = np.array(shock_none_data['hotelling']['rpdis'][firm1])
      array2 = np.array(shock_none_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

```

```

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the
    CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

array1 = array([0.57, 0.57, 0.57, 0.57, 0.57, 0.57, 0.57]), array2 =
array([0.49, 0.49, 0.49, 0.49, 0.49, 0.49, 0.49])

```

```

=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0800
Exact P-value:           0.01562
95% Test-Inverted CI:    Outside defined search grid
=====

```

4.2 Shock A

4.2.1 Average Prices

```
[19]: array1 = np.array(shock_a_data['hotelling']['avg_prices'][firm1])
      array2 = np.array(shock_a_data['hotelling']['avg_prices'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
      if len(ci_values) > 0:
          lower_bound = np.min(ci_values)
          upper_bound = np.max(ci_values)
          ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
      else:
          ci_str = "Outside defined search grid"
```

```
# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")
```

```
=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0100
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0100, 0.0100]
=====
```

4.2.2 Deltas

```
[20]: array1 = np.array(shock_a_data['hotelling']['deltas'][firm1])
      array2 = np.array(shock_a_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
```

```

        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the
    ↪ CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0014
Exact P-value:           1.00000
95% Test-Inverted CI:     [-0.0300, 0.0250]
=====

```

4.2.3 RPDIs

```

[21]: array1 = np.array(shock_a_data['hotelling']['rpdis'][firm1])
      array2 = np.array(shock_a_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪ dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

```

```

)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
EXACT PAIRED PERMUTATION TEST REPORT
=====

```

```

Observed Mean Difference: 0.0471
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0400, 0.0500]
=====

```

4.3 Shock B

4.3.1 Average Prices

```

[22]: array1 = np.array(shock_b_data['hotelling']['avg_prices'][firm1])
      array2 = np.array(shock_b_data['hotelling']['avg_prices'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
      if len(ci_values) > 0:
          lower_bound = np.min(ci_values)

```

```

upper_bound = np.max(ci_values)
ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0100
Exact P-value:           0.01562
95% Test-Inverted CI:     [0.0100, 0.0100]
=====

```

4.3.2 Deltas

```

[23]: array1 = np.array(shock_b_data['hotelling']['deltas'][firm1])
      array2 = np.array(shock_b_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta

```

```

shifted_array1 = array1 - shift

perm_res = permutation_test(
    data=(shifted_array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',
    alternative='two-sided',
    n_resamples=np.inf
)

# If p > 0.05, this shift magnitude is statistically plausible (inside the
CI)
if perm_res.pvalue > 0.05:
    ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0071
Exact P-value:           0.82812
95% Test-Inverted CI:     [-0.0900, 0.0700]
=====

```

4.3.3 RPDIs

```

[24]: array1 = np.array(shock_b_data['hotelling']['rpdis'][firm1])
      array2 = np.array(shock_b_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),

```

```

    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")

```

```
print("=====")
```

```
=====
      EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0614
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0600, 0.0650]
=====
```

4.4 Shock C

4.4.1 Average Prices

```
[25]: array1 = np.array(shock_c_data['hotelling']['avg_prices'][firm1])
      array2 = np.array(shock_c_data['hotelling']['avg_prices'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
```

```

        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0100
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0100, 0.0100]
=====

```

4.4.2 Deltas

```

[26]: array1 = np.array(shock_c_data['hotelling']['deltas'][firm1])
      array2 = np.array(shock_c_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)

```

```

ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪ CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0100
Exact P-value:           0.79688
95% Test-Inverted CI:    [-0.0700, 0.0850]
=====

```

4.4.3 RPDIs

```
[27]: array1 = np.array(shock_c_data['hotelling']['rpdis'][firm1])
      array2 = np.array(shock_c_data['hotelling']['rpdis'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
      if len(ci_values) > 0:
          lower_bound = np.min(ci_values)
          upper_bound = np.max(ci_values)
          ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
      else:
          ci_str = "Outside defined search grid"
```

```
# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")
```

```
=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0386
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0350, 0.0400]
=====
```

5 Linear

5.1 Shock None

5.1.1 Average Prices

```
[28]: array1 = np.array(shock_none_data['linear']['avg_prices'][firm1])
array2 = np.array(shock_none_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
```

```

        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪ CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0300
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.0300, -0.0300]
=====

```

5.1.2 Deltas

```

[29]: array1 = np.array(shock_none_data['linear']['deltas'][firm1])
      array2 = np.array(shock_none_data['linear']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪ dependencies

```

```

        alternative='two-sided',
        n_resamples=np.inf
    )

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

=====

EXACT PAIRED PERMUTATION TEST REPORT

```
=====
Observed Mean Difference: 0.0986
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0950, 0.1000]
=====
```

5.1.3 RPDIs

```
[30]: array1 = np.array(shock_none_data['linear']['rpdis'][firm1])
      array2 = np.array(shock_none_data['linear']['rpdis'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If p > 0.05, this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
      if len(ci_values) > 0:
          lower_bound = np.min(ci_values)
```

```

upper_bound = np.max(ci_values)
ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.3529
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.3600, -0.3500]
=====

```

5.2 Shock A

5.2.1 Average Prices

```

[31]: array1 = np.array(shock_a_data['linear']['avg_prices'][firm1])
      array2 = np.array(shock_a_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta

```

```

shifted_array1 = array1 - shift

perm_res = permutation_test(
    data=(shifted_array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',
    alternative='two-sided',
    n_resamples=np.inf
)

# If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the
CI)
if perm_res.pvalue > 0.05:
    ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0300
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.0300, -0.0300]
=====

```

5.2.2 Deltas

```

[32]: array1 = np.array(shock_a_data['linear']['deltas'][firm1])
      array2 = np.array(shock_a_data['linear']['deltas'][firm2])

      main_test = permutation_test(

```

```

    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the_
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:                {main_test.pvalue:.5f}")

```

```
print(f"95% Test-Inverted CI:      {ci_str}")
print("=====")
```

```
=====
EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0329
Exact P-value:           0.03125
95% Test-Inverted CI:    [-0.0550, -0.0150]
=====
```

5.2.3 RPDIs

```
[33]: array1 = np.array(shock_a_data['linear']['rpdis'][firm1])
array2 = np.array(shock_a_data['linear']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the_
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)
```

```

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.3671
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.3700, -0.3600]
=====

```

5.3 Shock B

5.3.1 Average Prices

```

[34]: array1 = np.array(shock_b_data['linear']['avg_prices'][firm1])
      array2 = np.array(shock_b_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)

```

```

ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0286
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.0300, -0.0250]
=====

```

5.3.2 Deltas

```
[35]: array1 = np.array(shock_b_data['linear']['deltas'][firm1])
      array2 = np.array(shock_b_data['linear']['deltas'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
          )

          # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the_
          ↪CI)
          if perm_res.pvalue > 0.05:
              ci_values.append(shift)

      # Extract boundaries from the valid grid values
      if len(ci_values) > 0:
          lower_bound = np.min(ci_values)
          upper_bound = np.max(ci_values)
          ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
      else:
          ci_str = "Outside defined search grid"
          print(f"{array1 = }, {array2 = }")
```

```
# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")
```

```
=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: 0.0886
Exact P-value:           0.01562
95% Test-Inverted CI:    [0.0200, 0.1700]
=====
```

5.3.3 RPDIs

```
[36]: array1 = np.array(shock_b_data['linear']['rpdis'][firm1])
      array2 = np.array(shock_b_data['linear']['rpdis'][firm2])

      main_test = permutation_test(
          data=(array1, array2),
          statistic=paired_mean_diff,
          permutation_type='samples',      # 'samples' handles paired data structural_
          ↪dependencies
          alternative='two-sided',
          n_resamples=np.inf
      )

      # --- 4. Compute the Test-Inverted 95% Confidence Interval ---
      # We run a fine grid search to map exactly where the p-value transitions past 0.
      ↪05
      shifts = np.linspace(-0.5, 0.5, 10001)
      ci_values = []

      for shift in shifts:
          # Shift sample x by a hypothetical effect size delta
          shifted_array1 = array1 - shift

          perm_res = permutation_test(
              data=(shifted_array1, array2),
              statistic=paired_mean_diff,
              permutation_type='samples',
              alternative='two-sided',
              n_resamples=np.inf
```

```

    )

    # If  $p > 0.05$ , this shift magnitude is statistically plausible (inside the
    ↪ CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.3514
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.3550, -0.3500]
=====

```

5.4 Shock C

5.4.1 Average Prices

```

[37]: array1 = np.array(shock_c_data['linear']['avg_prices'][firm1])
      array2 = np.array(shock_c_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural
    ↪ dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

```

```

)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    ↪CI
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
EXACT PAIRED PERMUTATION TEST REPORT
=====

```

```

Observed Mean Difference: -0.0300
Exact P-value:           0.01562
95% Test-Inverted CI:    [-0.0300, -0.0300]
=====

```

5.4.2 Deltas

```

[38]: array1 = np.array(shock_c_data['linear']['deltas'][firm1])
      array2 = np.array(shock_c_data['linear']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
    ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the_
    ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"

```

```

else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.0686
Exact P-value:           0.09375
95% Test-Inverted CI:    [-0.1450, 0.0100]
=====

```

5.4.3 RPDIs

```

[39]: array1 = np.array(shock_c_data['linear']['rpdis'][firm1])
      array2 = np.array(shock_c_data['linear']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural_
    ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(

```

```

        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
    ↪ CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")

# --- 5. Report Results ---
print("=====")
print("  EXACT PAIRED PERMUTATION TEST REPORT  ")
print("=====")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:           {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=====")

```

```

=====
  EXACT PAIRED PERMUTATION TEST REPORT
=====
Observed Mean Difference: -0.3686
Exact P-value:           0.01562
95% Test-Inverted CI:     [-0.3700, -0.3650]
=====

```